# Siting of Electric Vehicle Charging Stations Based on Weighted Voronoi Diagram: A Graphic User Interface Design

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***Abstract*—Electric vehicle charging station is the infrastructure to provide electric energy for electric vehicles (EV). Reasonable location of charging station is of great significance to promote the popularization of EVs. Firstly, the ownership of EVs in the selected area is forecasted. According to the ownership and driving characteristics of various types of EVs, the total charging demand of EVs and the charging demand of each network node are obtained. Secondly, the location model of charging station is established to minimize the sum of operation and maintenance costs of charging station and user loss costs. Finally, weighted Voronoi diagram and adaptive particle swarm optimization are used. The results show that this method can optimize the annual average comprehensive cost of charging station. Graphic user interface (GUI) is designed to display the result of location, which makes the location scheme more intuitive.**

***Index Terms*--Charging Demand; Charging Station Siting; Electric Vehicle; Weighted Voronoi Diagram; Graphic User Interface.**

## I. INTRODUCTION

Developing the EV industry is an important measure to reduce greenhouse gas emissions and reduce dependence on fossil fuels. With the development of EV technology and the strong support of some countries in policy, EV have developed rapidly in the past ten years [1]. However, there are still some common problems in the siting and layout of charging facilities. There are still many special problems to be solved in the construction and development of China [2].

EV charging stations can provide charging, maintenance and other services for EV, and are important facilities needed to develop the EV industry. Before studying the problem of location and capacity of EV charging stations, it is often necessary to first analyze the charging demand of EV [3], and the Markov chain [4] method can be applied to simulate the daily remaining power of EV [1], The Monte Carlo simulation method is used to predict the charging power of EVs [3]. Subsequent mathematical modeling of the location problem is usually aimed at minimizing the annual average cost of the charging facility [5], supplemented by a variety of other siting objectives as the overall objective function. The constraint conditions are generally EV charging demand constraints [6], charging station service radius constraints, traffic network traffic flow constraints [7], grid power quality and economic constraints and EV charging convenience constraints. In the process of solving the siting model, a variety of model solving algorithms are derived. Genetic algorithm is one of the most classical algorithms [8]. The immune algorithm is developed based on genetic algorithm, and the algorithm evaluates the individual more [9]. Comprehensive, and the choice is more reasonable. By simulating the case, the results verify that the above methods have their own characteristics and are applicable to different situations, depending on the requirements of solution quality, algorithm efficiency, problem size, algorithm properties, and system prerequisites [10]. However, in the above method, the algorithm is difficult to solve, and the complexity of the actual transportation network is not considered.

Aiming at the shortcomings of current charging station siting for EVs, a new charging station siting method is proposed. A charging station siting model considering the interests of users and operators is established. The model is solved by combining weighted Voronoi diagram and adaptive particle swarm optimization algorithm. The feasibility of the proposed method is verified by an example, and show the effect of the method through a GUI interface

## II. ELECTRIC VEHICLE CHARGING DEMAND FORECAST

### A. *Electric Vehicle Ownership*

Charging station siting should first determine the ownership of all types of EVs in the target siting year. According to the traditional vehicle ownership in the base year and the target siting year, combined with the Bass expansion model, it can be predicted. The discrete time formula of the model [11] is as follows:

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$$f(l+1)=P\left[1-F(l)\right]+QF(l)\left[1-F(l)\right] \tag{1}$$

Where $f(l+1)$ is the ratio of the number of newly purchased users to the number of potential users when $l+1$; $F(l)$ is the ratio of the total number of purchased users to the total number of potential users when it is $l$; $P$ is the innovation coefficient; $Q$ is the imitation coefficient. However, EVs are still in the promotion stage in China, and lack relevant data. Therefore, this paper analyzes the key parameters of the Bass model in the traditional automobile market to obtain the key parameters of the Bass model of EVs. The model parameters are expressed as:

$$\begin{cases} EV_P = CV_P \dfrac{S_{EV}}{S_{CV}} \\ EV_Q = CV_Q \dfrac{S_{EV}}{S_{CV}} \end{cases} \tag{2}$$

Where $EV_P$ and $EV_Q$ are the $P$ and $Q$ values of the Bass model of the EV respectively; $CV_P$ and $CV_Q$ are the $P$ and $Q$ values of the traditional automobile Bass model respectively; $S_{EV}$ is the internal and external influence score of the EV; $S_{CV}$ is the inside and outside of the traditional automobile impact score.

Assuming that the annual growth rate of traditional cars is 10%, and the potential number of EVs accounts for 80% of the annual increase in car sales, the number of EVs is

$$N_{ev}=0.8\Delta N_{cv} f(l+1) \tag{3}$$

Where $N_{ev}$ is the total annual planned quantity of EVs to the target; $\Delta N_{cv}$ is the annual growth of traditional vehicles.

### B. Electric Vehicle Charging Demand

#### 1) Electric vehicle total charging demand

The daily electricity demand of each type of EV is

$$P_a^b = S_a^t W_a^e \quad \forall a \in A \tag{4}$$

Where $a$ is the vehicle type; $A$ is the vehicle type set; $A=\{Pr, Ta, Bu\}$, which respectively represents the private car, taxi and bus; $P_a^b$ is the *b-th* electric car daily power in the *a-type* electric car demand; $S_a^t$ is the average daily driving distance of $a$ class of EVs; $W_a^e$ is the power consumption of $a$ class of EVs for 100 kilometers.

#### 2) Road network node charging requirements in the planned area

The service range of the EV charging station depends on the user's charging demand. In order to reduce the calculation dimension, the traffic flow of each section in the planned area is represented by the traffic flow of the road network node [12]. According to the traffic volume in the road network, and the charging station charging demand is distributed at the road network node to obtain the charging demand distribution of the road network node, that is

$$D_k = \frac{f_k}{\sum_{k\in N_D} f_k} L \quad \forall k \in N_D \tag{5}$$

Where $k$ is the charging demand point; $N_D$ is the charging demand point set; $D_k$ is the charging demand size of $k$ point; $f_k$ is the road segment traffic volume.

## III. Siting Model Based on Weighted Voronoi Diagram

### A. Weighted Voronoi Diagram

The Voronoi diagram, also known as the Thiessen polygon, can be seen as a pattern formed by expanding each vertex in the set of growth points p as a growth point to the same speed until it meets on a plane.

Assuming $p=(p_1, p_2, ..., p_n)$, $3\le n<\infty$ is a set of points on the Euclidean plane, then the conventional Voronoi diagram at any point is defined as

$$V(p_g)=\left\{x=V(p_g)\,\middle|\,d(x,p_g)\le d(x,p_h), h=1,2,...,n, g\ne h\right\} \tag{6}$$

The weighted Voronoi diagram is an extended form of the conventional Voronoi diagram [11], defined as

$$V(p_g,\lambda_g)=\left\{x\in V(p_g,\lambda_g)\,\middle|\,\frac{d(x,p_g)}{\lambda_g}\le\frac{d(x,p_h)}{\lambda_h}, h=1,2,...,n, g\ne h\right\} \tag{7}$$

Where $\lambda_g$ ($g$=1,2,...,n) is a given n positive real numbers; $d(x,p_g)$ and $d(x,p_h)$ are the Euclidean distances of point $x$ and $p_g$ and $p_h$; $\lambda_g$ is the weight of $p_g$, $p_g \ne p_h$, $g\ne h$, $g$, $h\in\{1,2,...,n\}$; $x$ is any point on the plane.

### B. Siting Model

Under the condition that it can meet the regional maximum EV charging demand and the charging station service radius, the mathematical model of charging station siting optimization is established with the goal of minimizing the total annual cost of operation to target.

$$\mathrm{Min} C = C_1 + C_2 + C_3 \tag{8}$$

$$C_1 = \frac{r_0(1+r_0)^m}{(1+r_0)^m-1}\sum_k (C_k^B + A_k C_k^L) \tag{9}$$

$$C_2 = \sum_k 365(\alpha+\beta)\varepsilon p N_k^{\text{car}} \tag{10}$$

$$C_3 = 365\theta\varepsilon \sum_k \sum_j d_{kj} \quad k=1,2,...,N_{\text{chr}}; j\in J_k \tag{11}$$

$$J_1 \cup J_2 \cup ... \cup J_{N_{\text{chr}}} = J \tag{12}$$

$$N_k^{\text{chr}} \gamma_{\max} \le N_k^c (k=1,2,...,N_{\text{chr}}) \tag{13}$$

$$\max(d_{kj}) \le D_k \quad k = 1, 2, ..., N_{\text{chr}}; j \in J_k \tag{14}$$

In the formula, $C_1$ is the charging station converted to the annual construction cost; $C_2$ is the annual running cost of the charging station; $C_3$ is the user charging driving cost caused by the user's charging behavior; $r_0$ is the discount rate; m is the operating period of the charging station; $C_k^B$ is the construction cost of the charging station $k$ including the transformer purchase cost, the charger purchase cost and other fixed costs; $A_k$ And $C_k^L$ are the land acquisition area and unit price of the charging station $k$ respectively; $\alpha$ and $\beta$ are the conversion rates of the charging power consumption rate, the charging station operation and maintenance cost, the personnel salary and the like, respectively; $\mathcal{E}$ is the average number of charging times per day for the EV; $p$ is each The average charging cost of the secondary charging; $N_k^{\text{car}}$ is the number of EVs within the service range of the charging station $k$; $\theta$ is the driving cost conversion coefficient of the EV user; $J_k$ is the charging demand set of the charging station $k$; $\gamma_{\max}$ is the maximum simultaneous charging rate, and $N_k^c$ is the charging station The number of chargers in $k$; $d_{kj}$ is the travel distance from the charging demand point $j$ to the target charging station $k$; $D_k$ is the maximum service radius of the charging station $k$.

## IV. Graphic User Interface Design

In order to facilitate the observation of the siting results, it is also possible to visually see the service range of each charging station, and design a graphical user interface (GUI) to assist the display of the results of the siting. The initial interface is shown in Fig. 1.

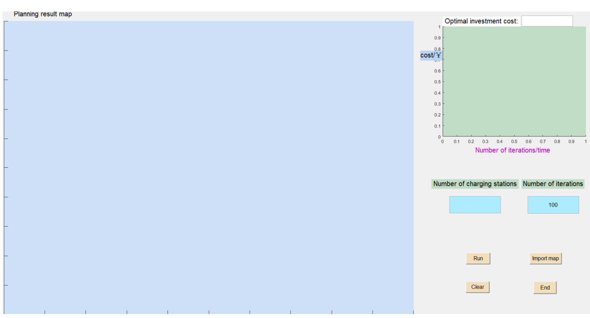


Figure 1. Graphic user interface diagram

In the GUI interface, the left side is the result of siting on the basis of the actual road network. The right side is divided into several parts, and the upper part is the optimal investment cost change curve after several iterations. In the middle are two parameter boxes that need to be input. One is the number of charging stations that plan to invest, and the other is the number of iteration calculations. The lower part is the four control buttons, which are to import the road network map, start running iteration, clear the iteration result and exit the GUI interface.

In the design of GUI system, the adaptive particle swarm optimization (APSO) algorithm [8] is used to solve the problem of target location. Adaptive particle swarm optimization is an improved particle swarm optimization algorithm (PSO). By modifying the inertia weight $\omega$ and the learning factors $c_1$ and $c_2$, the PSO algorithm can jump out of the local optimal position with greater probability when it falls into local optimum and improve the ability of global optimization.

## V. Case Analysis

### A. Overview of the Case

Take the siting of an EV charging station in Chengdu in 2025 as an example. The siting area is shown in the following figure. In order to simplify the calculation, the electric private car and taxi model are selected by the Emgrand EV300, and the electric bus model is BYD K9.

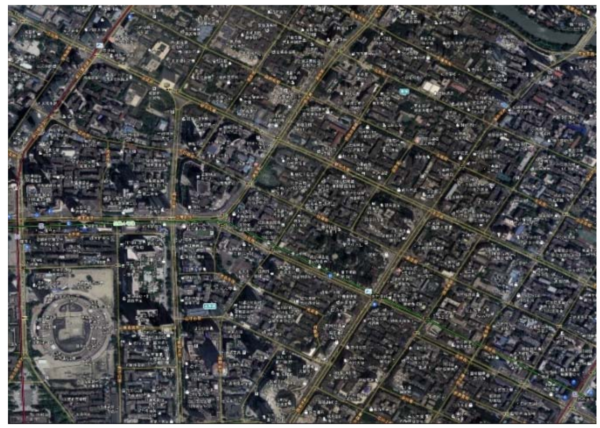

Figure 2. Siting area diagram

### B. Electric Vehicle Ownership Forecast

Based on 2015, the number of EVs in 2025 is predicted. The number of private cars, taxis and buses in the region in 2015 was 118,507, 49,375 and 26,019 respectively. The innovation coefficient $P$ of the traditional car Bass model parameters was 0.003, and the imitation coefficient $Q$ was 0.209. The benchmark of the traditional automobile market is set to 1. The results of scoring the internal and external factors of each type of EV are shown in Table below. Then, according to (2), the $P$ and $Q$ parameters of the Bass model of the EV are obtained.

According to the results of $P$ and $Q$, the predicted values of EV ownership are obtained from (1) and (3). By 2025, the number of electric private cars, taxis and buses is 53,106, 10,427 and 7,702 respectively.

TABLE I. Comparison of Influencing Factors between EVs and Conventional Cars

| Car type | External influence factor | | | Internal influence factor | |
|---|---|---|---|---|---|
| | Govern ment policy | Econom ic conditio n | Number of competit ors | Product quality | Product life |
| Traditio nal car | 1 | 1 | 1 | 1 | 1 |
| Electric private car | 1.6 | 0.5 | 0.2 | 1 | 1.1 |

| | | | | | |
|---|---|---|---|---|---|
| Electric taxi | 1.5 | 0.5 | 0.1 | 1 | 1 |
| Electric bus | 1.2 | 0.5 | 0.1 | 1 | 0.4 |

### C. *Siting Result*

According to the charging requirements of each node of the road network, the annual comprehensive cost of the number of different charging stations is shown in Table below.

TABLE II. ANNUAL COMPREHENSIVE COST OF DIFFERENT NUMBER OF CHARGING STATIONS

| Number of charging stations | Annual comprehensive cost / 10,000 yuan |
|---|---|
| 3 | 3161.31 |
| 4 | 2050.13 |
| 5 | 4068.66 |
| 6 | 6087.17 |
| 7 | 8105.70 |
| 8 | 10124.2 |
| 9 | 12142.8 |
| 10 | 14161.3 |

When the number of proposed charging stations is 4, the minimum annual comprehensive cost is 20.5113 million yuan. If the number of charging stations continues to increase, the annual comprehensive cost will gradually increase. And when the number of charging stations is less than 4, the annual comprehensive cost is larger than that of charging stations at 4. Therefore, according to economic considerations, it is most appropriate to plan 4 charging stations in this area. The iteration process is shown in Fig. 3. The number of iterations is 100.

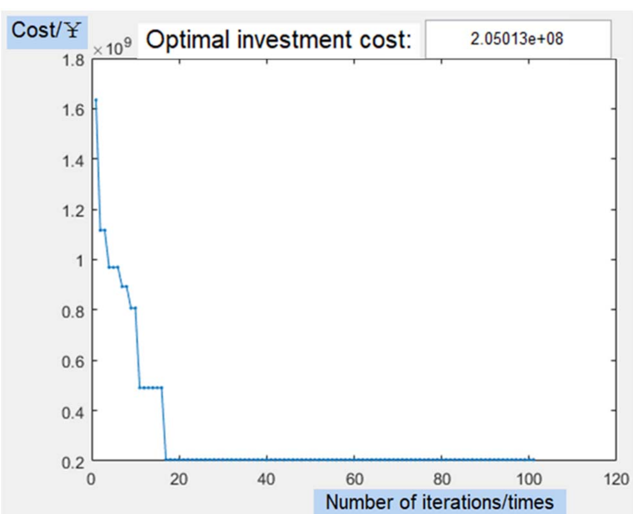


Figure 3. Iterative process of charging station number 4

In order to show the siting results, different siting schemes for the number of charging stations of 5, 6, and 7 are simulated, as shown in Fig. 4, Fig. 5, and Fig. 6.

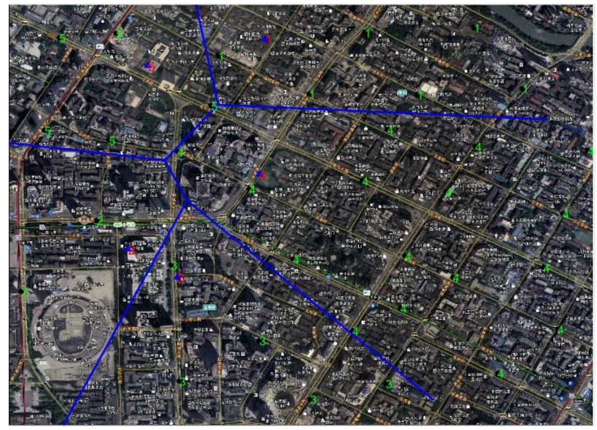

Figure 4. Siting scheme of charging station number 5

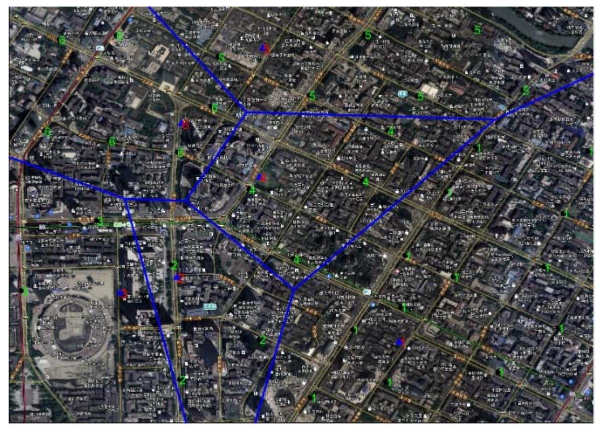

Figure 5. Siting scheme of charging station number 6

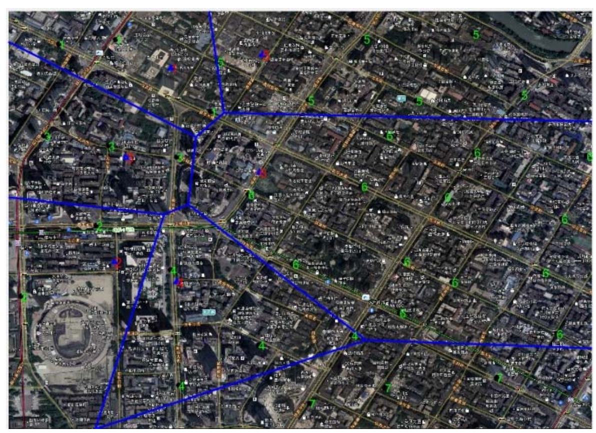

Figure 6. Siting scheme of charging station number 7

In the figure, the solid blue line indicates the siting range, that is, the service range of each charging station allocated by the Voronoi diagram; the red number represents the charging station number, that is, the location of the charging station in the area; and the green number indicates the road. The electric car on the network node can look for the charging station responsible for charging in the area. As can be seen from the above figure, when the number of planned charging stations is different, not only the annual comprehensive cost is different, but also the service range of the charging station will change accordingly.

As the number of charging stations increases, the service range of each charging station will decrease accordingly, which means that EVs in this area have more convenient and quick path selection to reach the nearest charging station to meet their charging needs, but at the same time The annual comprehensive cost will also increase accordingly, because more charging stations mean greater investment, and if the number of charging stations is too large, it may cause the load rate to decrease. Therefore, it is not possible to blindly build a new charging station. The balance between economy and convenience.

It is most economical to allocate the service range of each charging station as shown in the figure. After 100 iterations, the scheme with the lowest annual comprehensive cost is obtained. Other allocation schemes necessarily mean greater annual comprehensive cost and require more investment.

## VI. CONCLUSION

Aiming at the problem of EV power supply facility siting, this paper proposes a charging method for EV charging station, aiming at the minimum of the charging station construction cost, operation and maintenance cost and user loss cost to meet the charging station service capacity. The charging station position distance is constraint, the charging station siting model is established. The analysis of the example shows that the introduction of the particle swarm algorithm effectively avoids the disadvantages of using the Voronoi diagram to determine the site address, which makes the location of the charging station and the service scope more reasonable. In terms of economy, the siting plan is optimized with the minimum annual average cost, taking into account the interests of EV users and charging station operators. And use the designed graphical user interface to display the siting results, making the siting scheme more intuitive and convincing.

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